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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43206 Roll No. | | | | | | | | | | |

B. Tech. Examination, 2025-26

(Odd Semester)

DIGITAL LOGIC DESIGN

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : 8×1=8

   (a) What is the quine Mc-culsky method is used for?

   (b) Simplify using K-map :

   $$F(x, y, z) = \Sigma (1, 3, 5, 6, 7)$$

   (c) What is a hazard in a combinational logic circuits?

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(d) What is a half adder? Draw its logic diagram?

(e) What is the basic difference between latch and a flip flop?

(f) What is a ripple counter?

(g) What is race free state assignment?

(h) Explain the difference between PLA and PAL?

SECTION – B

2. Attempt any two parts of the following : 2×6=12

(a) Draw and explain the circuit diagram of magnitude comparator for two 4-bit numbers.

(b) What is the difference between demultiplexer and decoder? Implement the 8×1 multiplexer using the given function

$F(A, B, C, D) = \Sigma (1, 3, 4, 5, 8, 9, 11, 13, 15)$

take D as input.

(c) Explain and design the circuit of decade (mod 10) counter.

(d) Draw and explain serial in parallel out register.

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. 8×5=40

3. (a) Reduce the given Boolean function using K-map :

$F(w, x, y, z) = \Sigma (0, 2, 3, 4, 6, 7, 8, 12, 15) + d(w, x, y, z) = \Sigma (9, 13, 14)$

(b) Convert the given values as directed

(i) $(11010.111)_2 = (\ )_{10}$

(ii) $(2345)_{10} = (\ )_8$

(iii) $(567)_{10} = (\ )_{16}$

(iv) $(784.23)_{10} = (\ )_2$

(c) Reduce the given Boolean function using Tabular method :

$F(A, B, C, D) = \Sigma (0, 1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 15)$

4. (a) What is priority encoder explain with the help of circuit diagram and truth table?

(b) What are Hamming codes? What is the minimum hamming distance required to correct the one bit error? Generate a 7-bit hamming code for the data bits (1010) using odd parity.

(c) Design a PLA circuit for the given Boolean function

$F_1(x, y, z) = \Sigma(1, 3, 4, 5, 7), F_2 = \Sigma(0, 1, 2, 3, 4)$

5. (a) What are the various types of ROM explain in detail. Explain the working of PAL using block diagram.

(b) What are flip flops? Convert the SR flip-flop to T flip-flop?

(c) Design the 4-bit bidirectional shift register?

6. (a) An asynchronous sequential circuit has two internal state and one output .The excitation and output function describing the circuit are as follows :

$$Y_1 = x_1 x_2 + x_1 y_2' + x_2' y_1$$

$$Y_2 = x_2 + x_1 y_1' y_2 + x_1' y_1$$

$$Z = x_2 + y_1$$

(i) Draw the logic diagram of the circuit.

(ii) Derive the transition table and output map.

(iii) Obtain the flow table for the circuit.

(b) A synchronous sequential circuit has two flip flops one input x and one output y. The state table is shown. Design a circuit using D flip flop.

Present State	Next State		Output	
	x = 0	x = 1	x = 0	x = 1
AB	AB	AB	y	y
00	00	01	0	0
01	00	11	1	0
10	00	10	1	0
11	00	10	1	0

(c) Draw and explain the state diagram, state table, excitation table and logic circuit for 2-bit up counter.

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